

Please read the instructions to authors (https://academic.oup.com/nar/pages/Submission_Webserver) before submitting.

Proposal

[help](#) [example](#)

Short name	GRNsight <i>Please use a short name or an abbreviation, e.g. MyTool.</i>										
Title	GRNsight v7: a web application for visualizing models of small-scale gene regulatory and protein-protein interaction networks										
URL	https://dondi.github.io/GRNsight/ (including http:// or https://) Please check 'wget -v https://your.website/' if this does not work - errors are mainly caused by invalid SSL certificates.										
Free access	☒ There is a clearly visible statement or licence on the landing page granting free access to every users, including commercial users . ☒ Users can use this web server without email address.										
Login	☒ There is no login requirement. ☐ This website processes only sensitive human data (such as WGS or WTS or EHRs) and a login is required. ☐ This website <i>can</i> process sensitive data (such as ES or TS from humans) and a login is required for these. Access to non-sensitive data (e.g. other species or short DNA sequences) must be possible without login.										
Cookies	☐ This website does not use any cookies. ☐ This website uses session cookies and there is a clearly visible notification on the landing page. ☒ This website uses third-party cookies (e.g. youtube.com or baidu.com) and there is a cookie consent form describing the different cookies and why they are set.										
Updates	☒ This is an update of a webserver already published in the WSI or elsewhere. Pubmed ID(s): https://doi.org/10.7717/pe ☐ This is an update of a manuscript/proposal that was been rejected by the WSI. Running number: 034										
Method(s)	Please list the Pubmed IDs of the manuscripts in which your method was validated: https://doi.org/10.7717/pe We will not consider web-based classifiers unless the classifier was published in a peer-reviewed journal and the PMID is specified here.										
Type	☒ webserver ☐ web service Please submit stand-alone as a standard manuscript at https://mc.manuscriptcentral.com/nar (category: Computational Biology).										
External data	☒ I confirm that I am allowed to include all used external data in this application.										
Keywords help	gene regulatory network (GRN);protein-protein interaction network (PPI);graph visualization;time-course gene expression Search for keywords below (autocomplete list): Enter the first keyword here...										
Cover letter max. 300 words	GRNsight has been in continuous development since January 2014. We use our GitHub repository to practice open development where bugs and issues are tracked. We practice continuous integration using GitHub Actions which integrates workflows for automated builds, unit tests, and linting to streamline development and improve maintainability. GRNsight v1 with basic force graph layout and edge coloring functionality was first published in 2016 (Dahlquist, et al., PeerJ Computer Science, 2, e85. https://doi.org/10.7717/peerj-cs.85). Many new features described in the summary have been implemented afterwards up to the current v7. We originally published in PeerJ Computer Science because its open review process matched our open-source values. However, because our article was not indexed in PubMed or the Web of Science, the visibility of our software to biologists was limited, and the article has only been cited 4 times outside of our group. We monitor GRNsight usage via Google Analytics, which reported 7,536 unique users, 7,836 file uploads, 18,470 sessions, and 39,464 pageviews as of 2023. Our group has regularly presented posters and talks at professional conferences including Intelligent Systems for Molecular Biology, the Bioinformatics Open Source Conference, Yeast Genetics and Molecular Biology, Society for Women Engineers, and SoCal Systems Biology, and undergraduate conferences including the Southern California Conference for Undergraduate Research and the West Coast Biological Sciences Undergraduate Research Conference. We hope that inclusion in the 2026 NAR Web Server issue will raise the visibility of our software in the biological community. While the GRNsight database only contains budding yeast data at this time, GRNsight can display user-uploaded data from any species. And although originally designed for gene regulatory networks, we believe that GRNsight has general applicability for displaying any small, unweighted or weighted network with directed or undirected edges for systems biology or other application domains.										
One-page summary max. 300 words	GRNsight is an open-source web application for visualizing small-scale models of gene regulatory networks (GRNs) and protein-protein interaction networks (PPIs), optimized for 35 nodes and 70 edges or fewer. Network data in the form of an adjacency matrix (.xlsx), .sif, or .graphml format is either uploaded by the user or selected from our backend database derived from the Saccharomyces Genome Database. One-click demos and downloadable sample files are provided. GRNsight rapidly displays graphs using a physics-based force simulation via our customizations to the open source D3 JavaScript library. Genes/proteins are shown as labeled, rectangular nodes with Bezier-curved edges between them. Nodes can be rearranged manually or automatically laid out in a grid. GRNs have directed edges to indicate regulatory relationships, and PPIs have undirected edges to indicate physical binding. For GRNs, weighted edges are displayed with red pointed arrowheads for activation and blue blunt-end markers for repression; edge thickness indicates the magnitude of the regulatory relationship. Display of the numerical edge weights can be toggled on or off. Edge thicknesses are normalized to the largest magnitude weight value, and the normalization factor can be adjusted to allow fair visual comparison between different graphs. Edges with normalized weight values within 5% of 0 are displayed as gray to minimize their visual strength; this threshold can be adjusted with a slider. Gray edges can also be shown as dashed lines. Each node can display time-course expression data for that gene as an individual heat map, either uploaded by the user or selected from published yeast datasets stored in our backend database. Different expression data can be displayed across the top and bottom of a node to compare datasets. Graph images can be exported to .png, .svg, and .pdf formats, while graph data can be exported to .xlsx, .sif, or .graphml. The proposal must be submitted in the 'one-page summary' field. The proposal will be reviewed on the basis of the one-page summary entered above. You can add a PDF for figures. The PDF must not be longer than one page. Upload figures in a PDF: Browse... Dionisio-et-al_NAR-Web-Server_2026_image.pdf										
Browsers/OSs tested help	<table><thead><tr><th>OS</th><th>Chrome</th><th>Edge</th><th>Firefox</th><th>Safari</th></tr></thead><tbody><tr><td>Linux</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td></tr></tbody></table>	OS	Chrome	Edge	Firefox	Safari	Linux	☐	☐	☐	☐
OS	Chrome	Edge	Firefox	Safari							
Linux	☐	☐	☐	☐							